

June 19, 2026

Prof. Dr. Holger Boche
Institute of Theoretical Information Technology
Technical University of Munich (TUM)
Munich, Germany

Dear Prof. Boche and the Admissions Committee,

I am writing to express my strong interest in the Research Associate/PhD Student position focusing on the Information Theory of Secure Sensing and Coordination (ISSC). As an AI & Data Science Engineer with deep expertise in continuous-time dynamic systems, the challenge of securing robotic networks against eavesdropping via information-theoretic principles rather than computationally heavy cryptography is exactly the type of foundational research I wish to pursue.

In my Master's thesis, "Cardiovascular Medical Digital Twin," I engineered Medical_MambaTS, a linear-time state-space model capable of sub-millisecond bedside inference. This project taught me how to strictly bound highly chaotic, non-stationary temporal data. Furthermore, my work on PI-DeepONet, a Physics-Informed Operator Network, demonstrated my ability to enforce strict physical and mathematical constraints (such as 0D Windkessel fluid dynamics) directly within the neural network's architecture.

In the context of the ISSC project, this background in state-space representations translates directly to modeling dynamic coordination channels. I am particularly interested in tackling the problem of secure sensing using Physical Layer Security (PLS) and the Wyner Wiretap Channel model. By injecting Artificial Noise (AN) into the null-space of legitimate receivers, we can actively degrade an eavesdropper's signal-to-noise ratio, ensuring that the equivocation rate approaches the source entropy. This guarantees coordination stability independent of an adversary's quantum or classical computing capabilities.

I am confident that my mathematical rigor and practical engineering experience make me an excellent fit for the ACES Lab. I look forward to the opportunity to discuss my theoretical suggestions with you in an interview.

Thank you for your time and consideration.

Sincerely,
Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>